

System and Network Security

Dr. Ashok Kumar Das

IEEE Senior Member

Web of Science (Clarivate™) Highly Cited Researcher

Professor

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About me: Dr. Ashok Kumar Das

- **Education:** Ph.D. in Computer Science and Engineering, M.Tech. in Computer Science and M.Sc. in Mathematics and Computing from **IIT Kharagpur**, India
- **Visiting Research Professor** at the **Old Dominion University (ODU), Virginia, USA**, with the Virginia Modeling, Analysis, and Simulation Center (2022, 2024)
- **Adjunct Professor** at the **Korea University, Seoul, South Korea** from July 2024 onwards [2026 QS World Rank: 61]
- **Research Interests:** Network and system security, blockchain, security in Internet of Things (IoT), Internet of Vehicles (IoV), Internet of Drones (IoD), cloud/fog computing, intrusion detection, AI/ML security, and post-quantum cryptography
- **Research Highlights [Research Publications (Total: 526)]:**
 - ▶ Number of Journal Papers: 454
 - ▶ Number of Conference Papers: 59
 - ▶ Number of Book Chapters: 10
 - ▶ Number of Edited Books/Volumes: 3
 - ▶ Number of IEEE Transactions/IEEE Journal/IEEE Magazine Papers: 218

- **Total citations: 30,089, h-index: 98, i10-index: 359**
(According to Google Scholar Citations as on January 2, 2026)
- **Published in top venues like**
 - * IEEE Transactions on Information Forensics and Security
 - * IEEE Transactions on Dependable and Secure Computing
 - * IEEE Transactions on Consumer Electronics
 - * IEEE Transactions on Smart Grid
 - * IEEE Transactions on Industrial Informatics
 - * IEEE Transactions on Vehicular Technology
 - * IEEE Internet of Things Journal
 - * IEEE Consumer Electronics Magazine
 - * IEEE Communications Magazine
 - * IEEE Journal of Biomedical and Health Informatics
 - * IEEE Transactions on Network Science and Engineering
 - * IEEE Transactions on Intelligent Transportation Systems
 - * IEEE Sensors Journal
 - * IEEE Transactions on Network and Service Management
 - * IEEE Communications Surveys and Tutorials
 - * ACM Computing Surveys
 - * ACM Transactions on Sensor Networks

About me: Journal Editorial Board Members

- Associate Editor: **IEEE Transactions on Information Forensics and Security** (SCI Impact Factor: 6.8)
- Associate Editor: **IEEE Systems Journal** (SCI Impact Factor: 4.4)
- Editor: **Journal of Network and Computer Applications** (Elsevier) (SCI Impact Factor: 8.7)
- Editor (Technical Committee): **Computer Communications** (Elsevier) journal (SCI Impact Factor: 6)
- Associate Editor: **Journal of Cloud Computing** (Springer) (SCI Impact Factor: 4)
- Associate Editor: **Alexandria Engineering Journal** (Elsevier) (SCI Impact Factor: 6.8)
- Associate Editor: **IET Communications** journal (SCI Impact Factor: 1.6)
- Associate Editor: **Cyber Security and Applications** (Elsevier)
- Editor: **KSII Transactions on Internet and Information Systems** (SCI Indexed Journal)
- Editor: **International Journal of Communication Systems** (Wiley) (2024 -) (SCI Impact Factor: 1.7)
- Guest Editor: **IEEE Transactions on Consumer Electronics**, 2025
- Guest Editor: **IEEE Journal of Biomedical and Health Informatics**, 2026
- Guest Editor: **Computers & Electrical Engineering** (Elsevier) (SCI Indexed Journal), 2016
- Guest Editor: **ICT Express** (Elsevier) (SCI Indexed Journal), 2019
- Guest Editor: **Wireless Communications and Mobile Computing** (SCI Indexed Journal), 2020

About me

- Invited speaker at the **World Laureates Summit**, Dubai, UAE, 3-5 February 2026, in the area of Datascience and Cryptography, and Security, organized by the World Laureates Association (WLA) and the World Governments Summit Organization (WGS), in collaboration with IIT Madras Global Research Foundation.
- Mentor – IIM Lucknow Enterprise Incubation Centre (CoE–Blockchain Technology), supported by the Government of Uttar Pradesh, India, under the StartInUP Policy and IIML-EIC onboarded Microsoft as the Technology Partner for CoE-BT, 2026.
- Member of Panel Discussion: Security by Design: Embedding Trust, Ethics, and Transparency into AI Systems, at Cybersecurity and Privacy Workshop, 18th IEEE International Conference on COMMunication Systems & NETworks (COMSNETS 2026), January 6 - 10, 2026, Bangalore, India.
- Member of the Technical Program Committeess (TPCs) for **MAHA-AI for Science & Engineering (AI-SE) Mission of Anusandhan National Research Foundation (ANRF), Department of Science & Technology**, New Delhi, Government of India.
- Visiting Research Professor at the **Old Dominion University (ODU)**, Suffolk, VA 23435, USA, with the Virginia Modeling, Analysis, and Simulation Center (VMASC) [2022, 2024]
- **Distinguished Adjunct Professor** at the **Korea University, Seoul, South Korea** [2025 QS World Rank: 67], for two-year term starting from July 2024
- Listed in the **Web of Science (Clarivate™) Highly Cited Researcher 2022, 2023** in recognition of exceptional research performance demonstrated by production of multiple highly cited papers that rank in the top 1% for field and year.
- Included in the **top 2% scientists world-wide (all fields) and also in the area of Networking & Telecommunications, with Rank world-wide (by subject area): 39 for the year: 2022 (October 2023).**
- 2025–2022 - Research.com Computer Science in India Leader Awards

- **Mauro Conti**, IEEE Fellow, Head of SPRITZ Security and Privacy Research Group, Director of UniPD node of CINI Cybersecurity National Lab, EU Marie Curie Fellow Alumni, CEO and co-funder of CHISITO, and Co-funder of DYALOGHI, University of Padua, Italy
- **Willy Susilo**, IEEE Fellow, ARC Future Fellow, Co-Director, Centre for Computer and Information Security Research, University of Wollongong, Australia
- **Sajal K. Das**, IEEE Fellow, Professor and Daniel St. Clair Endowed Chair, Department of Computer Science, Missouri University of Science and Technology, Rolla, MO 65409, USA
- **Sherali Zeadally**, Fellow of the British Computer Society and the Institution of Engineering Technology, Stevenage, U.K., University of Kentucky, Lexington, KY 405 06 USA
- **Kim-Kwang Raymond Choo**, Fellow, Australian Computer Society, Department of Information Systems and Cyber Security, The University of Texas at San Antonio, San Antonio, TX 78249 USA
- **Xinyi Huang**, Fujian Provincial Key Laboratory of Network Security and Cryptology, Fujian Normal University, Fuzhou, Fujian, China
- **Alexey Vinel**, Halmstad University, Halmstad, Sweden
- **Muhammad Khurram Khan**, FIET (UK), FBCS (UK), Center of Excellence in Information Assurance, King Saud University, Riyadh, Saudi Arabia

- **Athanasios V. Vasilakos**, Lulea University of Technology, Sweden
- **Minho Jo**, Department of Computer and Information Science, Korea University, Seoul, South Korea
- **Laurence T. Yang**, St. Francis Xavier University, Canada
- **Joel J. P. C. Rodrigues**, IEEE Fellow, National Institute of Telecommunications - Inatel, Brazil
- **Debiao He**, School of Cyber Science and Engineering, Wuhan University, Wuhan 430 072, China
- **Jong-Hyouk Lee**, Sangmyung University, Republic of Korea
- **Kee-Young Yoo**, Kyungpook National University, Daegu, Korea
- **Qi Jiang**, Xidian University, China
- **Sachin Shetty**, Old Dominion University, USA
- **Xiong Li**, Hunan University of Science and Technology, China
- **Mamoun Alazab**, Charles Darwin University, Australia
- **Mohammad S. Obaidat**, IEEE Fellow, University of Sharjah
- **YoungHo Park**, School of Electronics Engineering, Kyungpook National University, South Korea
- **Shantanu Pal**, School of Information Technology, Deakin University, Melbourne, Australia

Welcome to System and Network Security

● Network Security

- ▶ Basics of cryptography, Types of attacks, Passive and Active attacks.
- ▶ Overview of symmetric cryptography
- ▶ Overview of public-key cryptography
- ▶ Overview of Digital Signature Standards
- ▶ Key Management
- ▶ Encrypting communication channels: LLE and EEE.
- ▶ Authentication Applications: Kerberos, X.509 authentication service.
- ▶ Security at the Application Layer: PGP, S/MIME, Password Management.
- ▶ Security at the Transport Layer/Web Security: Web security considerations, Secure sockets layer and Transport Layer security, Secure Electronics Transaction.
- ▶ Security at the Network Layer: IPSec
- ▶ Security at the Datalink Layer: Internal and External Error Controls.

- **System Security**

- ▶ Intruders: Intruders, Intrusion detection, Intrusion prevention.
- ▶ Malicious Software: Virus and related threats, Virus countermeasures.
- ▶ Software Vulnerabilities: Phishing, Buffer overflow (BOF), Heap overflow, Format string attacks, Cross-site scripting (XSS), SQL Injection.

- **Advanced research topics:**

- ▶ Post-quantum security protocols for Internet-of-Things applications

- Books

- ▶ William Stallings, "Cryptography and Network Security: Principles and Practices," Pearson Education, 6th Edition.
- ▶ Bernard Menezes, "Network Security and Cryptography," Cengage Learning.
- ▶ Behrouz A. Forouzan, "Cryptography and Network Security," Special Indian Edition.

- Research papers

- ▶ IEEE Transactions
- ▶ ACM Transactions
- ▶ Elsevier
- ▶ Springer
- ▶ Wiley

Grading Method:: Relative

- Mid Sem: 25% (Closed books and notes exam)
- End Sem: 35% (Closed books and notes exam)
- Quizzes (2): 15%
- Lab Assignments (5): 25%

Examinations

- Mid-Semester Examination [MSE] (Syllabus: Upto Mid-Sem lectures)
- End-Semester Examination [ESE] (Syllabus: After Mid-Sem part)

Mathematically, $MSE \cap ESE = \emptyset$

All lecture materials will be made available at course moodle:

<https://courses.iiit.ac.in/>
under the “Resources” directory.

- NO compromise in Institute (IIIT Hyderabad) policy
- Attendance is **COMPULSORY**
- NOT permitted to leave any lectures after the attendance
- NOT allowed to attend lectures after 10 minutes
- Grades will be reduced **AUTOMATICALLY** by the academic office according to the attendance policy (For more details, consult the academic office)

Thank You!!!

Any Doubts?